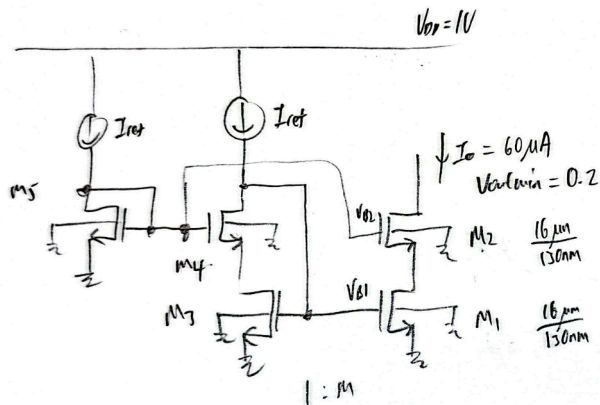


Chai Zi Yang A0262349Y EE5507 Tut 2

1. Hand Calculations



$$I_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right) V_{ov}^2$$

$$V_{ov} = \sqrt{\frac{60 \mu A}{\frac{1}{2} \times 100 \mu A/V^2 \times \frac{16 \mu m}{130 nm}}}$$

$$\approx 0.099 V$$

$\Rightarrow$ $V_{out,min}$ of 2V is already satisfied by M_1, M_2 sizing given

$$V_{THN} = 0.4V \quad \mu n C_{ox} = k_n' = 100 \mu A/V^2 \quad \lambda n = 0.61 V^{-1}$$

$$W, L > 130 nm \quad 2I_{ref} < 5 \mu A$$

Let $I_{ref} = 2.4 \mu A$ (total = $4.8 \mu A < 5 \mu A$)

$$M = \frac{60 \mu A}{2.4 \mu A} = 25$$

$$\Rightarrow (W/L)_3 = (W/L)_4 = \frac{16 \mu m}{130 nm} \div 25 \approx \frac{640 nm}{130 nm} \leftarrow \text{Keep } L \text{ the same so } \lambda_1 \approx \lambda_3 \text{ to deduce error}$$

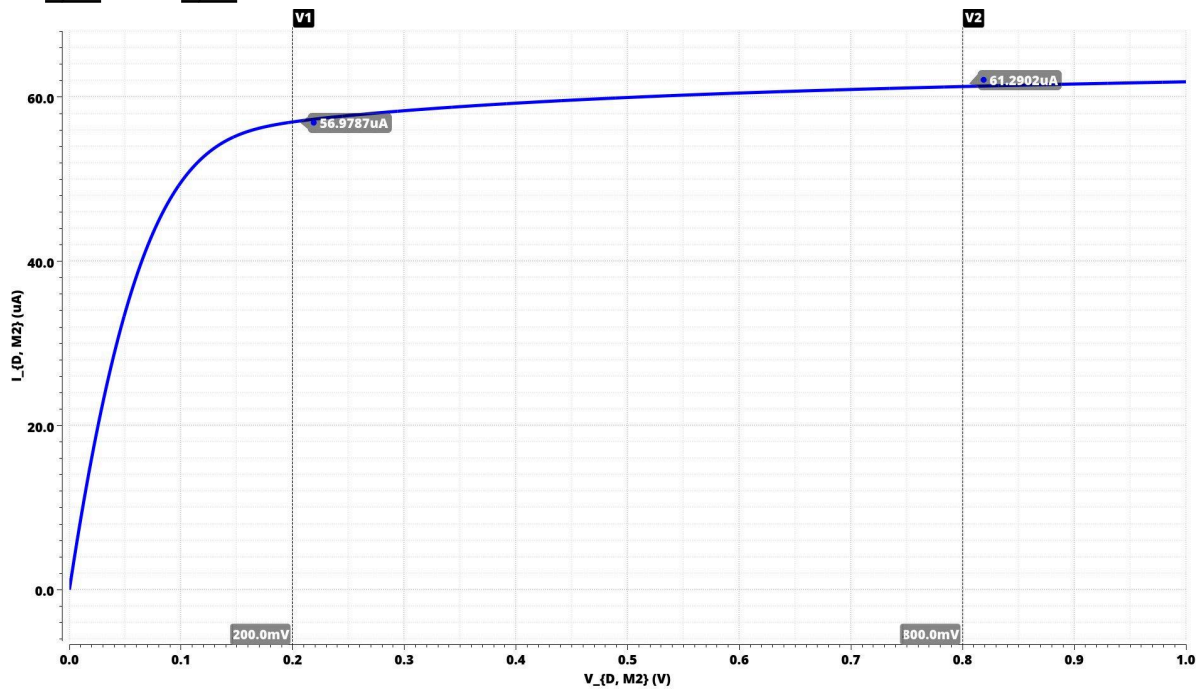
High swing cascode I $\Rightarrow (W/L)_5 = \frac{1}{4} (W/L)_3$

$$= \frac{160 nm}{130 nm}$$

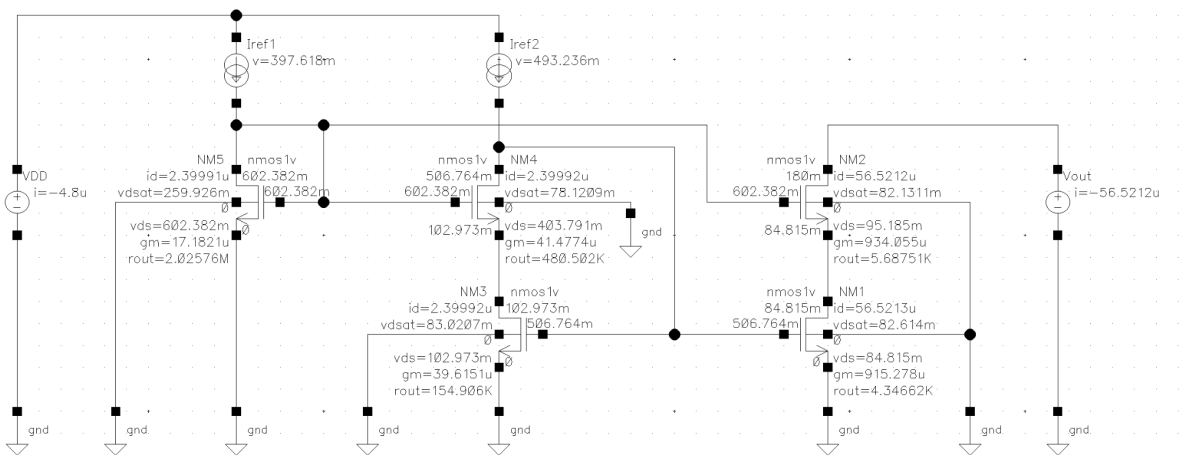
Let M_5 be standalone (since there is no point in making it the unit transistor) & let $M_3 = M_4$ be the unit

$$\Rightarrow (W/L)_5 = \frac{160 nm}{130 nm} \quad , \quad (W/L)_3 = (W/L)_4 = \frac{640 nm}{130 nm} \quad , \quad (W/L)_1 = (W/L)_2 = 25 \times \frac{640 nm}{130 nm}$$

2. $I_{D, M2}$ vs $V_{D, M2}$

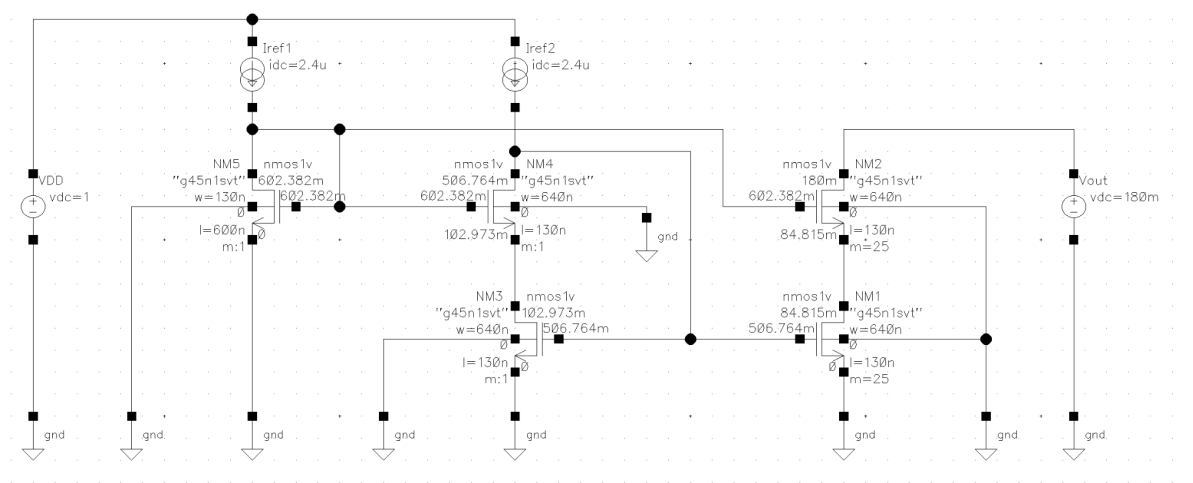


3. Final Schematic Showing DC Operating Point



DC operating point shown here is the worse case V_{out} with margin of 0.02 V, $V_{out} = 0.18 V$.

4. Final Schematic Showing Sizing



Final schematic defer with hand calculation only by the $(W/L)_{NM5}$. $(W/L)_{NM5}$ was reduced while I_{ref} was maintained at 2.4 μA to increase V_{GS5} and bring NM1/NM2 out of the linear region. In the end, $(W/L)_{NM5}$ was chosen to be 130nm/600nm to ensure NM1 and NM2 are both in the saturation region even under the case of $V_{out} = 0.18 V$.